Answer Key

Grade 6 Mathematics · Unit 3: Ratios and Rates · Answers for every item on Sheets 1 and 2 of each lesson packet. Constructed responses show a model answer; accept equivalent reasoning.

LESSON 3-1

Understanding Ratios · Ratios and Tape Diagrams

Sheet 1 · Understanding Ratios

1. a. 2 : 3 b. 3 : 2 c. 6 : 15 = 2 : 5 d. 9 : 15 = 3 : 5
2. a. 2 : 3 b. 5 : 2 c. 3 : 7
3. Part-to-whole. The 15 counts all the players on the team (8 girls + 7 boys).
4. Aquarium votes: 24 − 14 = 10. Museum to aquarium = 14 : 10 = 7 : 5. For every 7 museum votes, there are 5 aquarium votes.
5. Marcus reversed the order: 2 : 5 is cats to dogs (4 : 10). Dogs to cats is 10 : 4 = 5 : 2.

Sheet 2 · Ratios and Tape Diagrams

1. a. 3 + 5 = 8 boxes; 48 ÷ 8 = 6 beads b. Red: 3 × 6 = 18; Blue: 5 × 6 = 30 (18 + 30 = 48)
2. a. 18 and 24 b. 12 and 54
3. a. 12 ÷ 3 = 4 b. 4 × 4 = 16 melon pieces c. 12 + 16 = 28 pieces
4. a. Theo has 7 − 2 = 5 more boxes, so 5 boxes = 35 and one box = 7 stickers b. Maya 2 × 7 = 14; Theo 7 × 7 = 49; total 14 + 49 = 63 stickers
5. He divided 45 by 5 (one term) instead of by the total parts, so his shares add to 81, not 45. Total parts = 4 + 5 = 9; 45 ÷ 9 = 5 per box; shares are 4 × 5 = 20 and 5 × 5 = 25.

TEST PRACTICE

6. D – $25
7. B – 8

LESSON 3-2

Understanding Unit Rates · Comparing Unit Prices (Best Buy)

Sheet 1 · Understanding Unit Rates

1. a. 45 mi per hour b. $8 per ticket c. 70 words per minute d. 12 apples per bag
2. 1 h: $12; 2 h: $24; 4 h: $48. Unit rate: $12 per hour (36 ÷ 3 = 12).
3. 156 ÷ 6 = 26 smoothies per hour; 26 × 9 = 234 smoothies in 9 hours
4. a. 12 ÷ 4 = 3 cups per loaf b. 4 ÷ 12 = 1/3 loaf per cup
5. Ben divided in the wrong order; 0.02 is minutes per word. Words per minute = 150 ÷ 3 = 50 words per minute.

Sheet 2 · Comparing Unit Prices (Best Buy)

1. a. $1.50 per lb b. $0.50 per box
2. Small: $3.60 ÷ 12 = $0.30 per oz; Large: $5.04 ÷ 18 = $0.28 per oz. The large box is the better buy.
3. a. $46 ÷ 2 = $23 per gallon b. $107.50 ÷ 5 = $21.50 per gallon c. $23 − $21.50 = $1.50 per gallon
4. a. A: $5.00 ÷ 4 = $1.25; B: $7.80 ÷ 6 = $1.30 b. Store A ($1.25 < $1.30)
5. No. 32-oz jar: $5.12 ÷ 32 = $0.16 per oz; 16-oz jar: $2.40 ÷ 16 = $0.15 per oz. The smaller jar is the better buy.

TEST PRACTICE

6. B – $1.40 per lb
7. A – $0.45

LESSON 3-3

Equivalent Ratios and Tables · The Bridge Strategy (Scaling Ratios)

Sheet 1 · Equivalent Ratios and Tables

1. Nuts: 2, 4, 6, 10, 14; Cereal: 5, 10, 15, 25, 35
2. a. 20 b. 36 c. 4 d. 14
3. A, B, D (4 : 6 ÷ 2 = 2 : 3; × 2 = 8 : 12; 2 : 3 × 5 = 10 : 15; 6 : 8 adds 2 to each term)
4. 9 × 4 = 36, so 4 × 4 = 16 gold beads (table: 4 : 9, 8 : 18, 12 : 27, 16 : 36)
5. a. 3 × 7 = 21, so 2 × 7 = 14 cups of white b. 21 + 14 = 35 cups
6. No. Adding 3 to both terms does not keep the ratio; you must multiply or divide both terms by the same number. Example: 5 : 8 = 10 : 16 (× 2).

Sheet 2 · The Bridge Strategy (Scaling Ratios)

1. 2 minutes: 5 bottles (÷ 3); 10 minutes: 25 bottles (× 5)
2. a. 15 b. 8
3. a. 4 : 30 ÷ 2 = 2 : 15 b. 2 : 15 × 5 = 10 : 75, so 75 km
4. a. 10 : 4 ÷ 2 = 5 : 2 b. 5 : 2 × 3 = 15 : 6, so 15 cups
5. Adding 9 to both terms does not keep the ratio. Bridge: 6 : 10 ÷ 2 = 3 : 5, then × 5 = 15 : 25. She needs 25 spoons of water.

TEST PRACTICE

6. A – 10
7. A – 6 : 9 = 2 : 3 = 10 : 15

LESSON 3-4

Graphing Ratios on the Coordinate Plane · Interpreting Points on Ratio Graphs

Sheet 1 · Graphing Ratios on the Coordinate Plane

1. Oranges: 3, 6, 9, 12; pairs (1, 3), (2, 6), (3, 9), (4, 12)
2. Points (1, 3), (2, 6), (3, 9), (4, 12) lie on a straight line through (0, 0). (0, 0) means 0 bags hold 0 oranges.
3. a. (7, 14) b. (9, 18)
4. a. (8, 40): 8 × 5 = 40 plants b. (0, 0): 0 rows have 0 plants

Sheet 2 · Interpreting Points on Ratio Graphs

1. a. In 2 hours the cyclist rides 24 miles. b. 12 miles per hour c. 6 × 12 = 72 miles
2. a. 48 b. 5 c. 84 d. 3 e. 120 f. 8
3. a. 10 ÷ 4 = $2.50 per pound b. (6, 15): 15 ÷ 6 = 2.5, yes. (8, 18): 18 ÷ 8 = 2.25, no (8 lb would cost $20).
4. x is laps, so (3, 12) means 3 laps take 12 minutes (unit

5. No. (1, 5) and (2, 10) show 5 per 1, but 3 × 5 = 15, not 14. The point (3, 14) is not on the line.
6. She switched x and y. Tickets go first: 6 tickets cost 6 × $2 = $12, so the point is (6, 12).

LESSON 3-5

Comparing Ratios with Common Terms · Comparing Rates with Benchmarks

Sheet 1 · Comparing Ratios with Common Terms

1. a. 40 b. 42 c. 44 d. 27
2. P yellow: 8, 12, 16, 20; Q yellow: 12, 18, 24, 30. At 15 cups blue, P has 20 yellow and Q has 18 yellow, so Mix Q is bluer.
3. a. 9 : 12 × 4 = 36 : 48 b. 13 : 16 × 3 = 39 : 48 c. Pack B (39 of 48 sprout)
4. No. Scale to 12 spoons sugar: X = 12 : 27, Y = 12 : 30. X has less water for the same sugar, so Drink X is sweeter.

Sheet 2 · Comparing Rates with Benchmarks

1. a. 12 b. 18 (any common multiple is correct)
2. Kai: 90 × 3 = 270 words; Lena: 140 × 2 = 280 words. Lena types faster.
3. a. A: $8 × 8 = $64; B: $12 × 5 = $60 b. Store B
4. a. C: $7 × 4 = $28; D: $9 × 3 = $27 b. Store D
5. Benchmark 40 gallons: Car 1 goes 150 × 8 = 1,200 miles; Car 2 goes 200 × 5 = 1,000 miles. Car 1 is more efficient.

TEST PRACTICE

6. B – 24 bagels
7. B – Ella takes 180 minutes; Dante takes 200

LESSON 3-6

Customary Unit Conversions · Metric Unit Conversions

Sheet 1 · Customary Unit Conversions

1. a. 96 in b. 7 yd c. 80 oz d. 3 gal e. 14 cups f. 2.5 lb
2. a. < (3 ft = 36 in) b. < (5 pt = 2.5 qt) c. > (2 lb = 32 oz) d. = (3 yd = 9 ft)
3. 3 gal = 12 qt; 20 qt = 5 gal; 7.5 gal = 30 qt
4. 3 gal = 12 qt = 24 pt = 48 cups
5. No. 2 yd = 2 × 3 = 6 ft = 6 × 12 = 72 in. 70 in is less than 72 in, so she is 2 inches short.

Sheet 2 · Metric Unit Conversions

1. a. 3,500 m b. 0.85 m c. 6,000 mL d. 0.45 kg
2. a. 18 × 1,000 = 18,000 mL b. 18,000 ÷ 300 = 60 cups
3. a. 4 × 800 = 3,200 m b. 3,200 ÷ 1,000 = 3.2 km
4. a. 3 × 400 = 1,200 g b. No. 1.5 kg = 1,500 g; 1,500 − 1,200 = 300 g short
5. He divided by 1,000 instead of multiplying. Kilograms to grams is large to small, so multiply: 4.5 × 1,000 = 4,500 g.

TEST PRACTICE

6. B – 160 mm
7. C – 600 mL

LESSON 3-7

Customary vs. Metric Conversions · Multi-Step System Conversions and Dosage

Sheet 1 · Customary vs. Metric Conversions

1. a. 15.24 cm b. 40 km c. 5 kg d. 11.4 L e. 110 lb f. 5 mi g. 2 kg h. 16 km
2. 5 kg ≈ 11 lb; 22 lb ≈ 10 kg; 25 kg ≈ 55 lb
3. 96 ÷ 1.6 = 60; about 60 miles
4. a. 55 ÷ 2.2 = 25; about 25 kg b. No. 25 kg > 23 kg, so it is about 2 kg over the limit
5. 100 m ≈ 100 × 1.09 = 109 yd, which is more than 100 yd. The 100-meter race is longer (by about 9 yards).

Sheet 2 · Multi-Step System Conversions and Dosage

1. a. 35 kg b. 140 mg c. 6 tsp d. 1,500 mL
2. a. 8.8 ÷ 2.2 = 4 kg b. 4 × 5 = 20 mg
3. a. 4 × 400 × 5 = 8,000 m b. 8 km c. 8 ÷ 1.6 = 5 miles
4. a. 2 × 5 = 10; about 10 mL b. 120 ÷ 10 = 12; about 12 doses
5. She used pounds, but the rate is per kilogram. 66 ÷ 2.2 = 30 kg; 30 × 3 = 90 mg.

TEST PRACTICE

6. A – 33 ÷ 2.2 = 15 kg; 15 × 2 = 30 mg
7. A – 2.85 L